Cheat sheet

GenAI System Design

Framework, components and trade-offs for design rounds

Framework

1 Requirements — 2 Metrics — 3 Architecture — 4 Data and retrieval —
5 Model and prompts — 6 Evaluation and safety — 7 Scale, latency, cost

Clarifying questions

Users and scale	Who, how many, queries per second at peak
Latency	Time-to-first-token and total targets
Accuracy	Tolerance for errors; need for citations
Data	Sources, freshness, sensitivity, languages, permissions
Budget	Cost per query, infrastructure limits

Components

Gateway	Auth, rate limiting, quotas
Orchestrator	Prompting, tool calls, retries, routing
Retrieval	Ingestion pipeline, hybrid index, reranker, ACL filters
Models	Router between small and large; fallbacks
Guardrails	Input/output moderation, PII, injection defence
Caching	Exact and semantic caches; embedding cache
Observability	Traces, metrics, evaluation pipelines, feedback

Back-of-envelope

Tokens per query	Prompt + context + output (e.g. 3,000 in + 400 out)
Daily cost	Queries × tokens × price per token
Throughput	QPS × tokens per request ÷ tokens per second per replica

Pro tip

Always close with risks and how you would measure success after launch.